



Lecture 4: Regression

DD2421

Atsuto Maki

Autumn, 2026

Part I: we will visit

Regression => Real-valued output

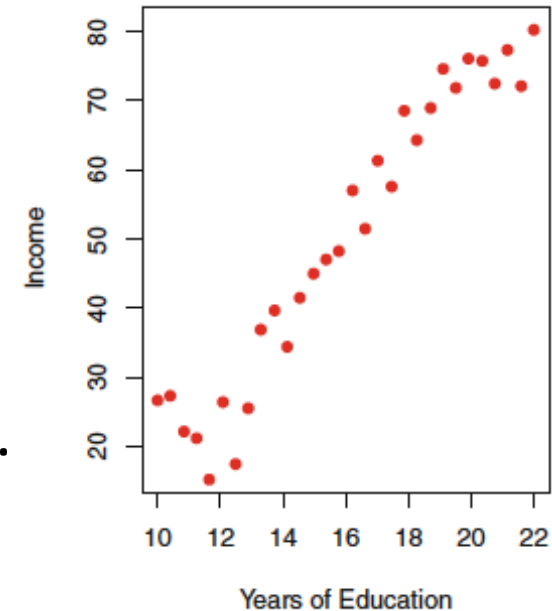
- Function approximation
- Linear Regression / Least Squares
 - Robust regression (RANSAC to handle outliers)
- Linear regression + regularization
 - Ridge regression
 - The Lasso (a more recent alternative)
- k -NN Regression

Function approximation

- How do we fit this dataset D ?

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$$

of N pairs of **inputs** x_i and **targets** $y_i \in R$.
 D can be measurements in an experiment.



- Task of regression:
to **predict** target associated to any arbitrary **new input**

Note: Here we have a single **input feature**, but inputs to regression tasks are often vectors $\mathbf{x}$ of **multiple input features (=attributes/predictors/variables)**.

Linear Regression (parametric)

Linear regression tries to estimate the function $f(x)$ (x is d -dimensional) and predict the output by

$$\hat{f}(x) = \sum_{i=0}^d w_i x_i = w^T x$$

How to measure the error for N samples:

- To see how well $\hat{f}(x)$ approximates $f(x)$, square error is used: $(\hat{f}(x) - f(x))^2$
- Mean Square Error: $E_{in}(\hat{f}) = \frac{1}{N} \sum_{n=1}^N (\hat{f}(x_n) - y_n)^2$ (“in-sample”)

Note: *error* in the context of minimization is often called *cost/loss/objective/energy*.

Minimizing in-sample MSE

E_{in} can be expressed as:

$$E_{in}(w) = \frac{1}{N} \sum_{n=1}^N (w^T x_n - y_n)^2 = \frac{1}{N} \|Xw - Y\|^2$$

where

$$X = \begin{bmatrix} x_1^T \\ \vdots \\ x_N^T \end{bmatrix}, \quad Y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

We want to compute the parameters w that minimize E_{in} .

Residual sum of squares (RSS)

The sum of squared errors is a **convex function** of w

$$E_{in}(w) = \|Xw - Y\|^2$$

The gradient with respect to the weights is:

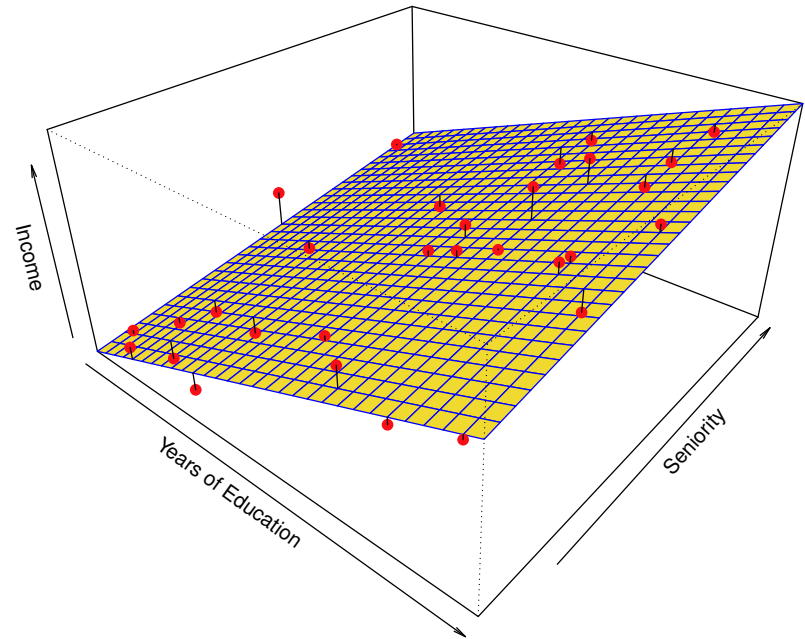
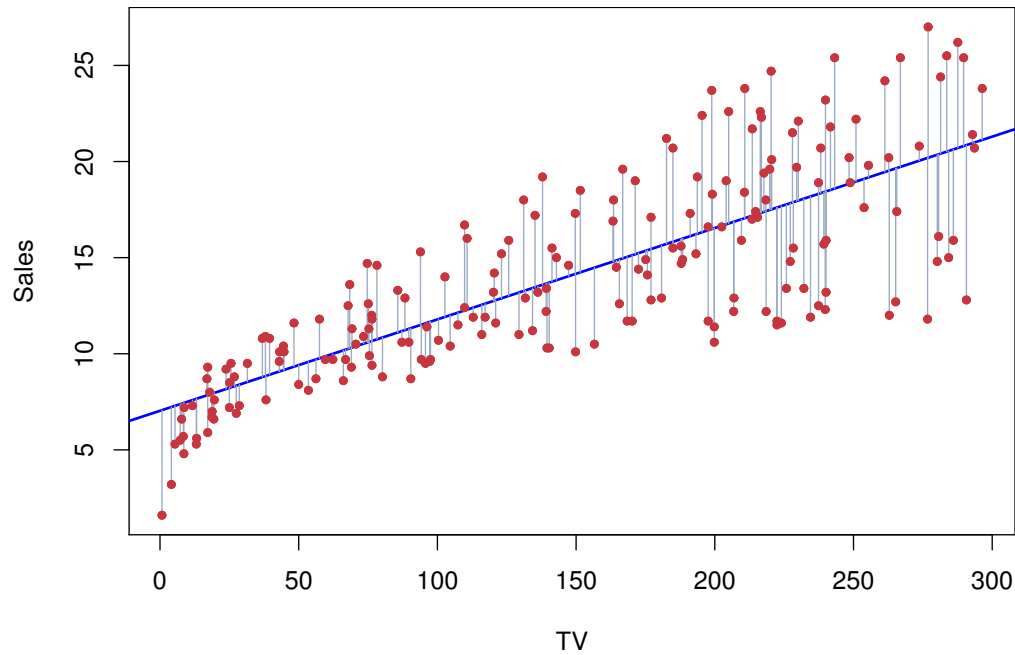
$$\frac{\partial}{\partial w} E_{in}(w) = 2X^T (Xw - Y)$$

The weight vector that sets **the gradient to zero** minimizes the errors

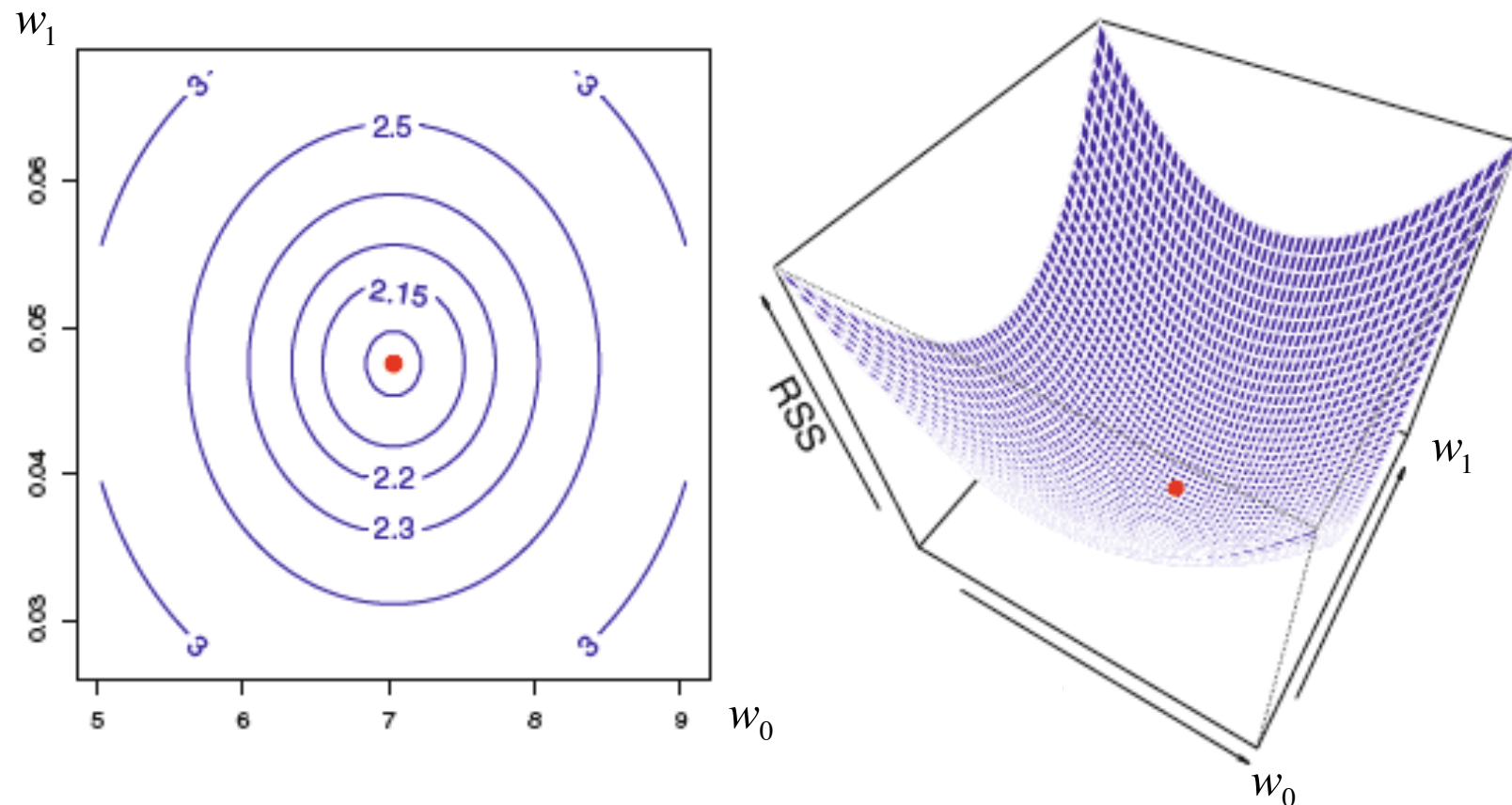
$$X^T Xw = X^T Y$$

$$w = (X^T X)^{-1} X^T Y$$

Examples of least squares fit



Examples of plots of RSS



Figures adapted from An Introduction to Statistical Learning (G. James et al.)

RANSAC: RANdom SAmpling Consensus

Robust regression

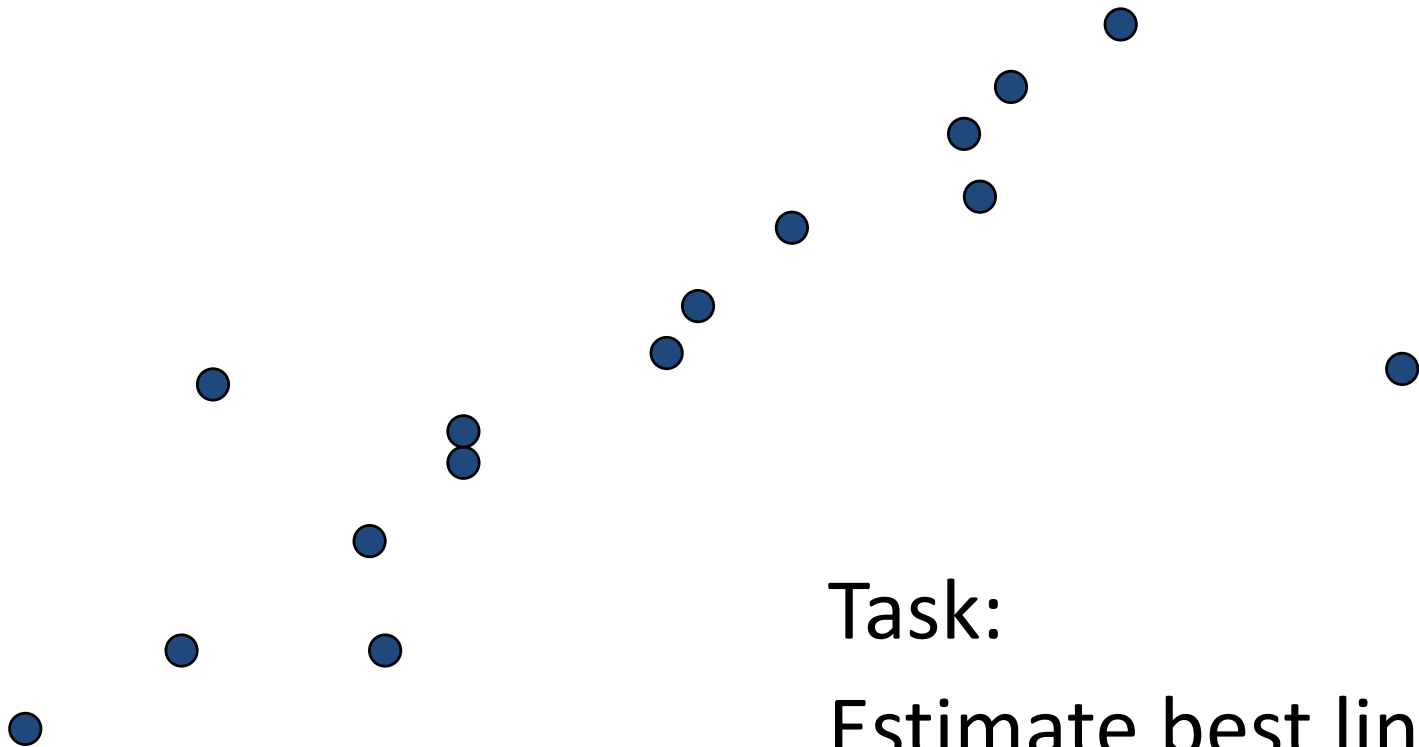
Repeat M times:

- Sample two points to estimate the line.
- Calculate the number of inliers or posterior likelihood for relation.
- Choose relation to maximize the number of inliers.

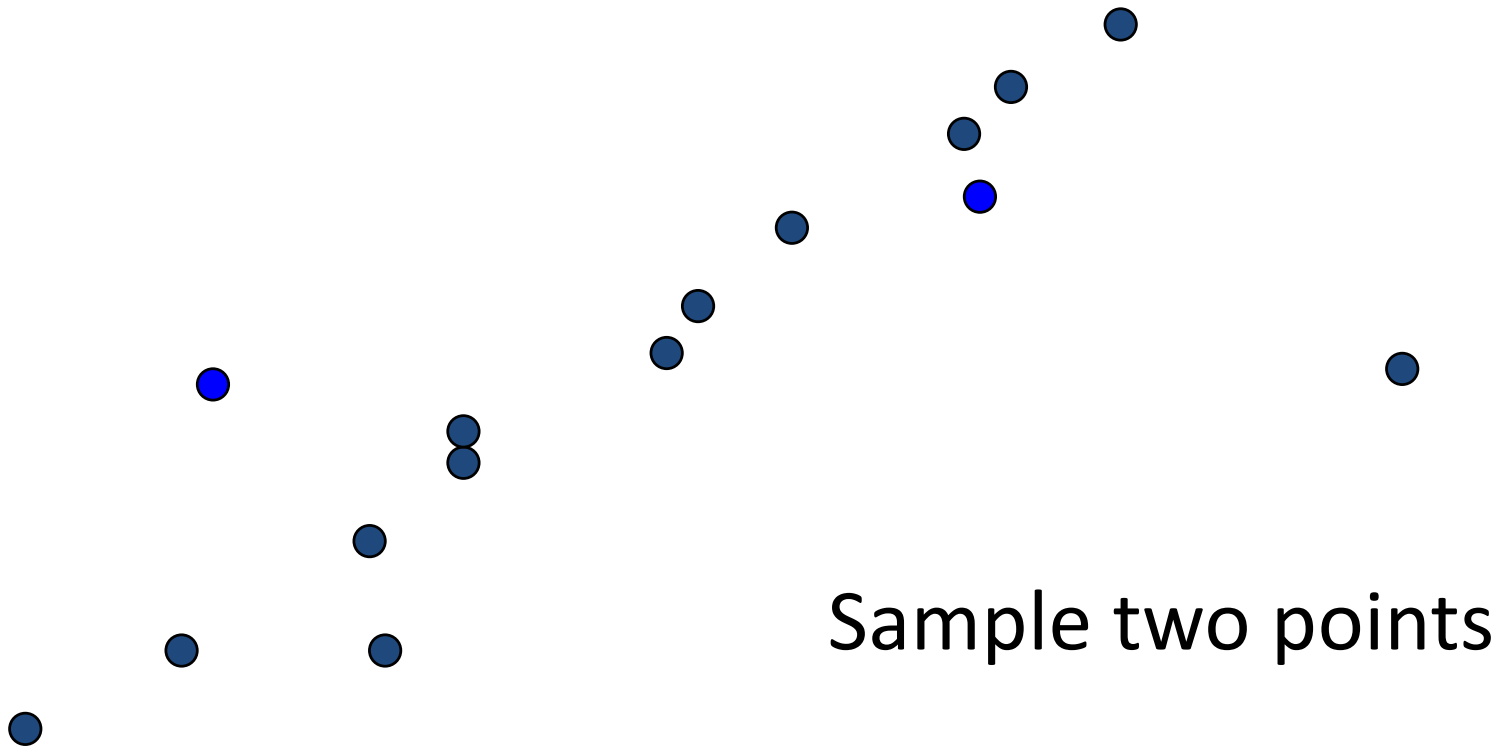
(Last two in *Least Median of Square* (LMedS))

- Calculate error of all data.
- Choose relation to minimize median of errors.

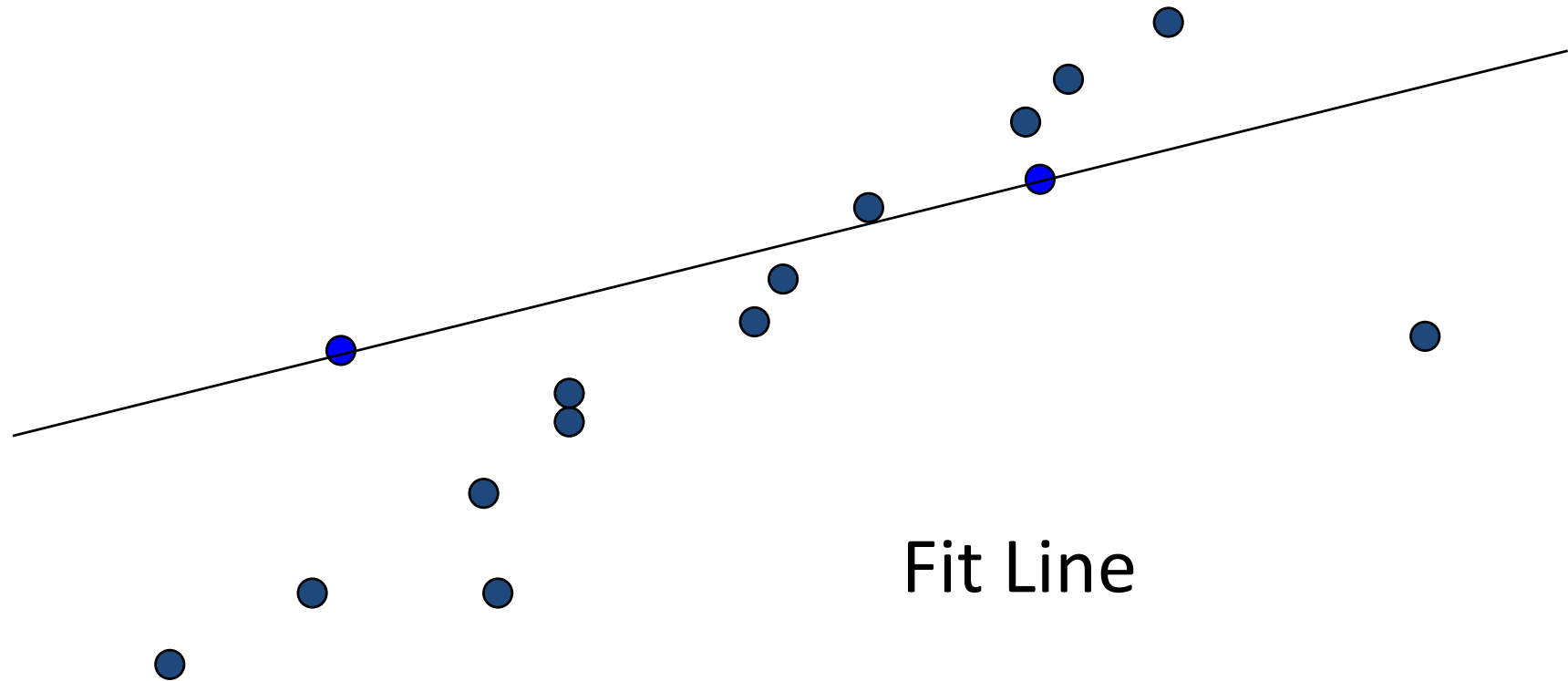
RANSAC line fitting example



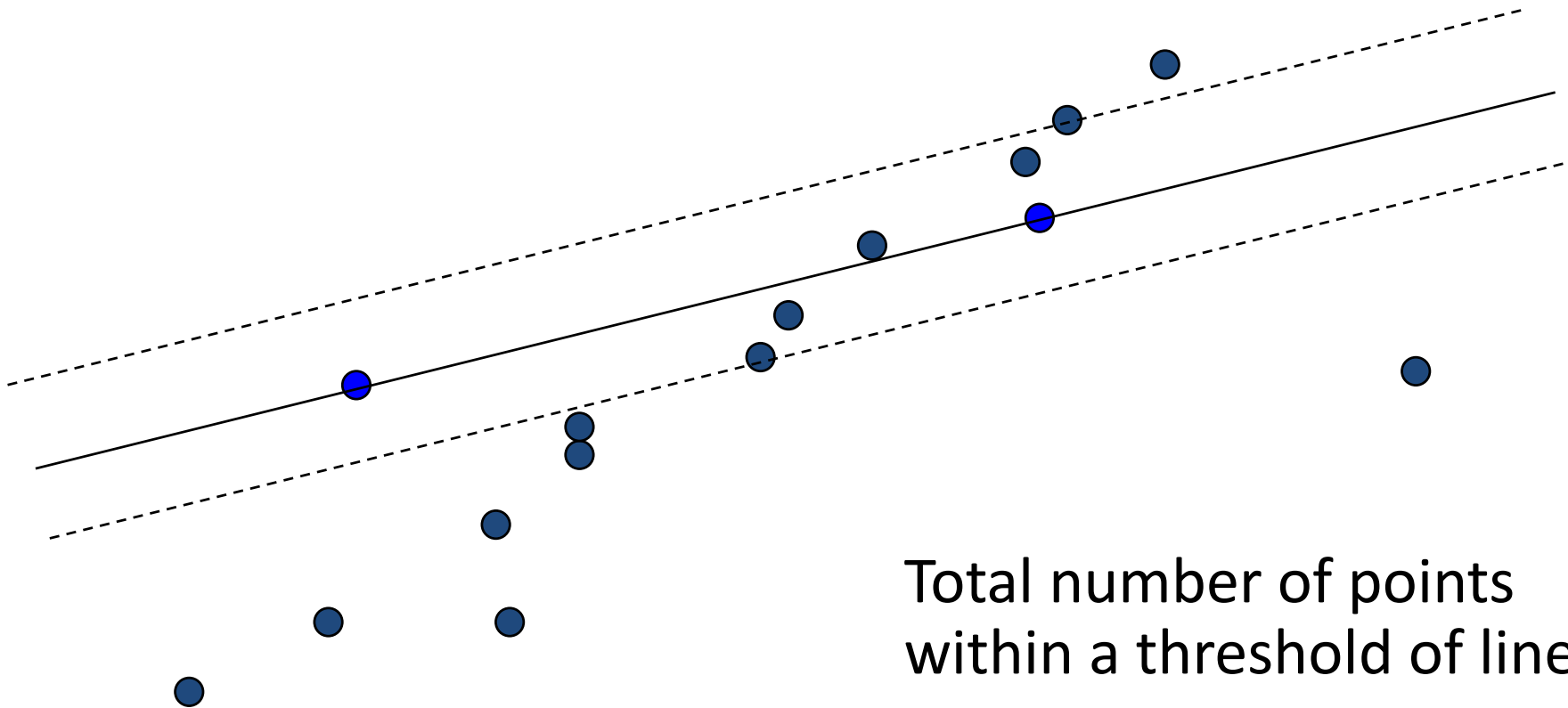
RANSAC line fitting example



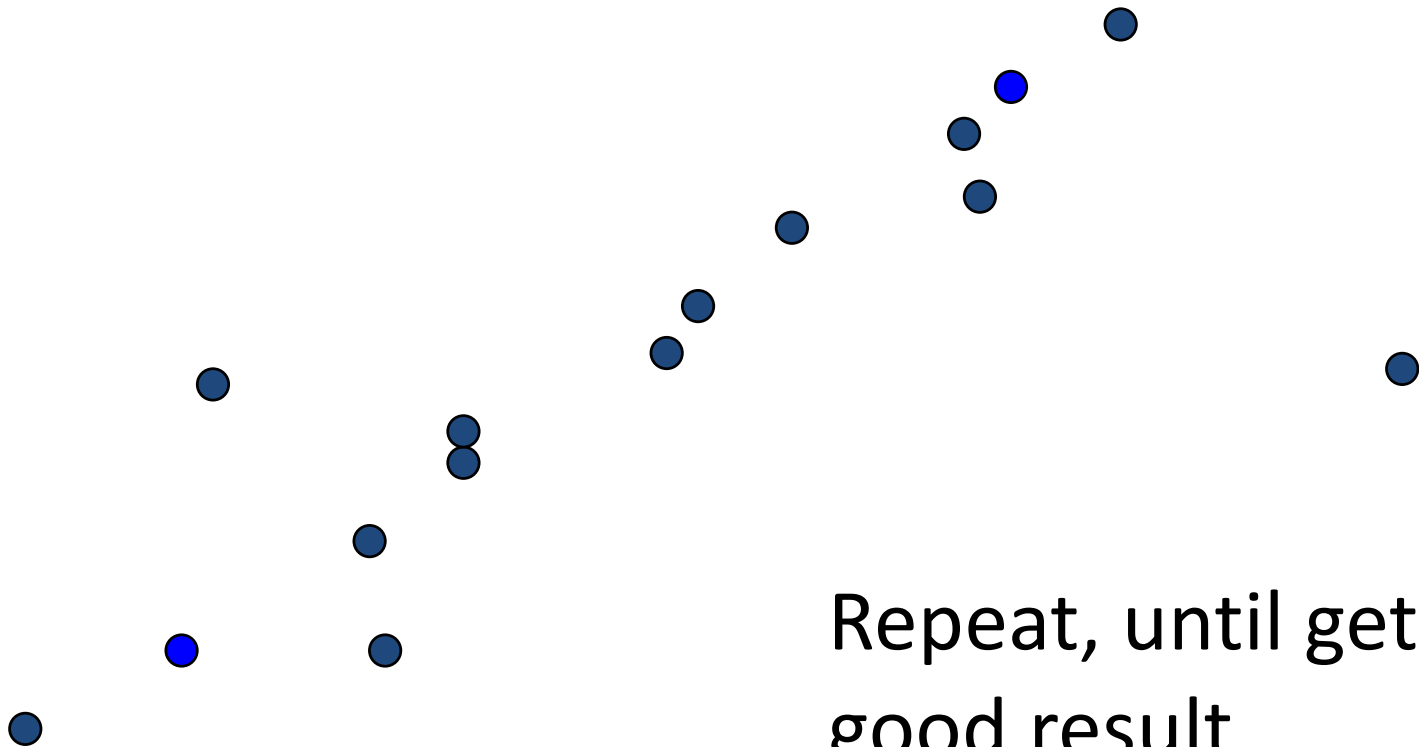
RANSAC line fitting example



RANSAC line fitting example

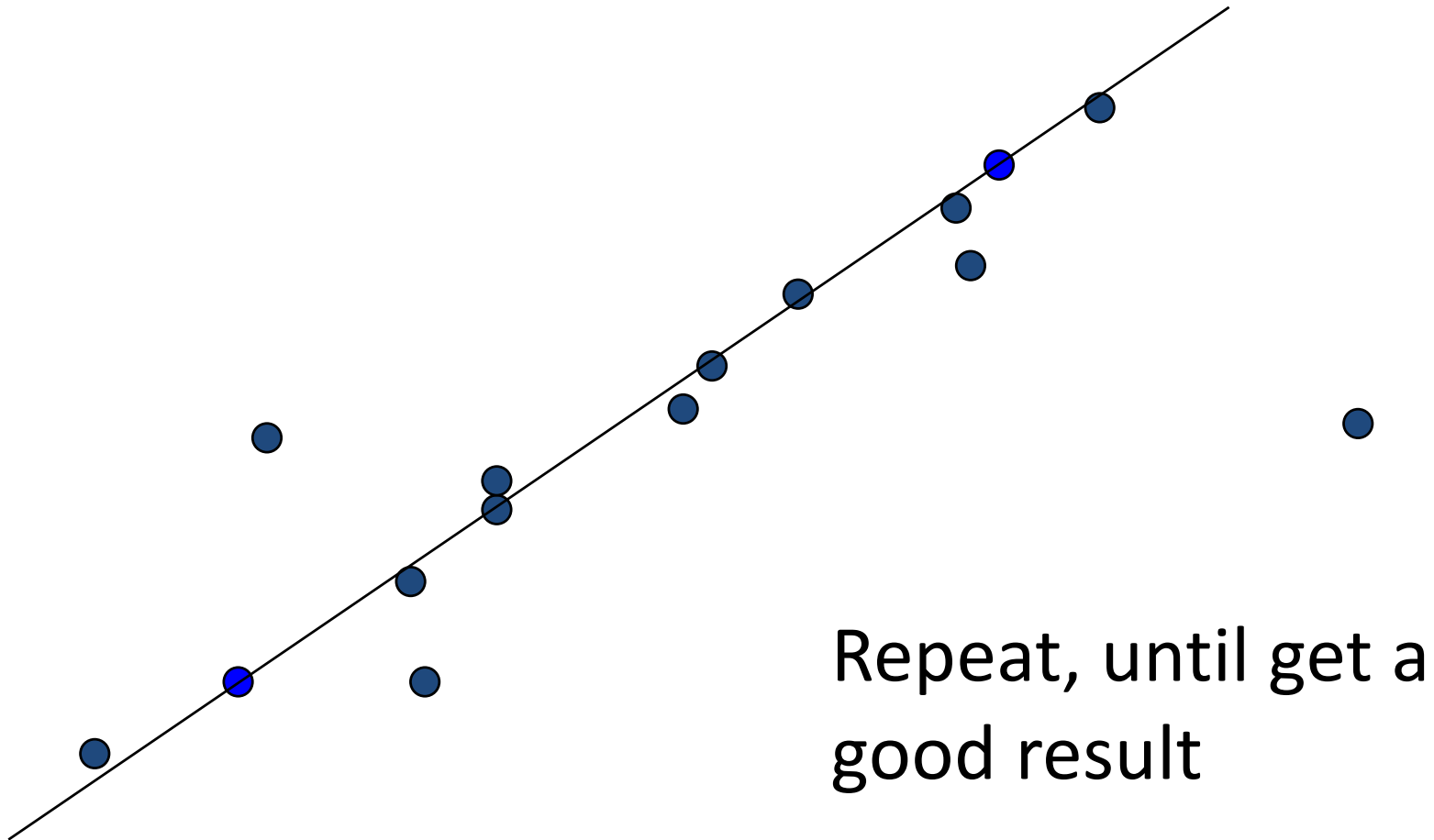


RANSAC line fitting example

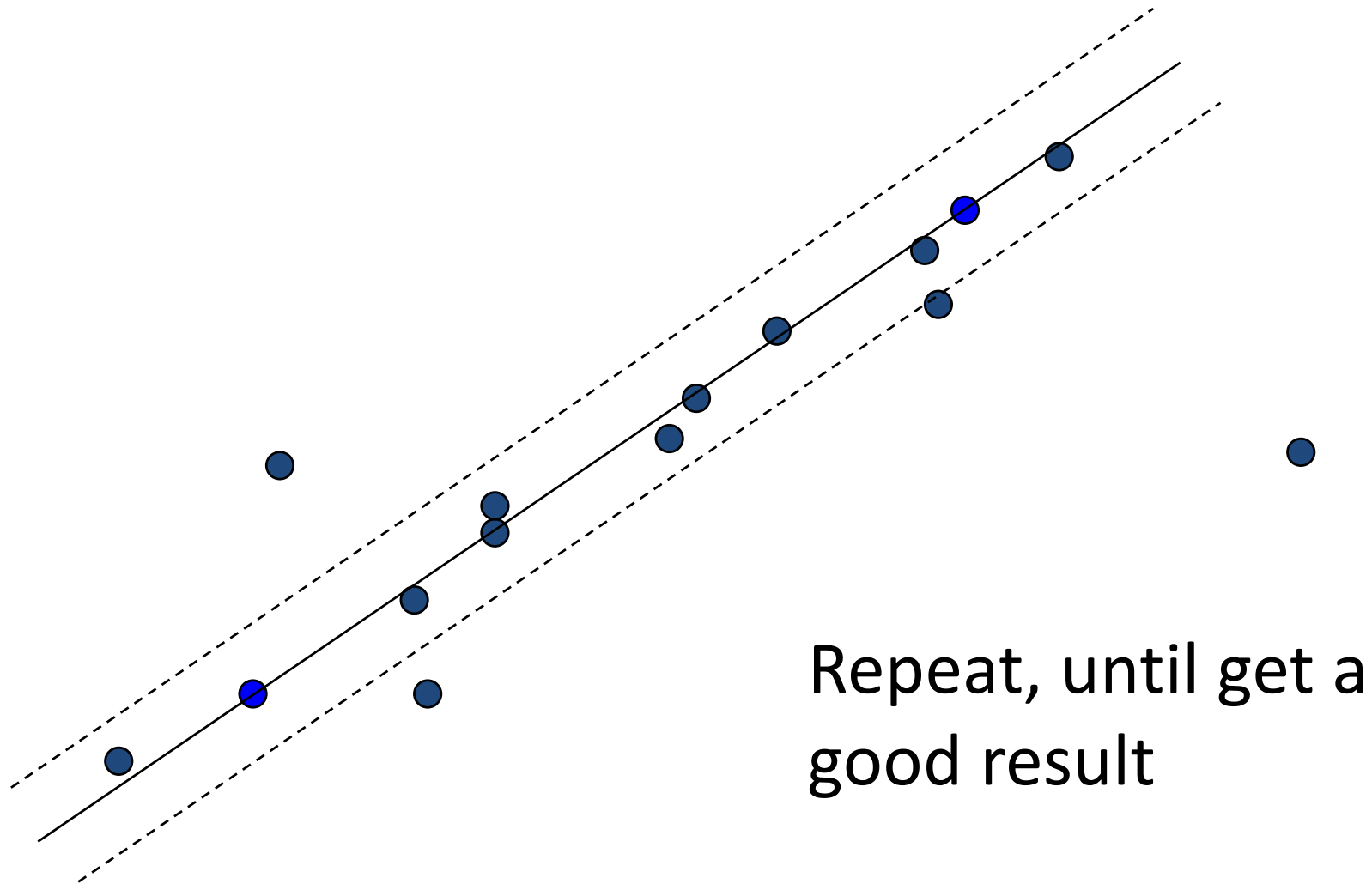


Repeat, until get a good result

RANSAC line fitting example



RANSAC line fitting example



RANSAC: RANdom SAmpling Consensus

Objective

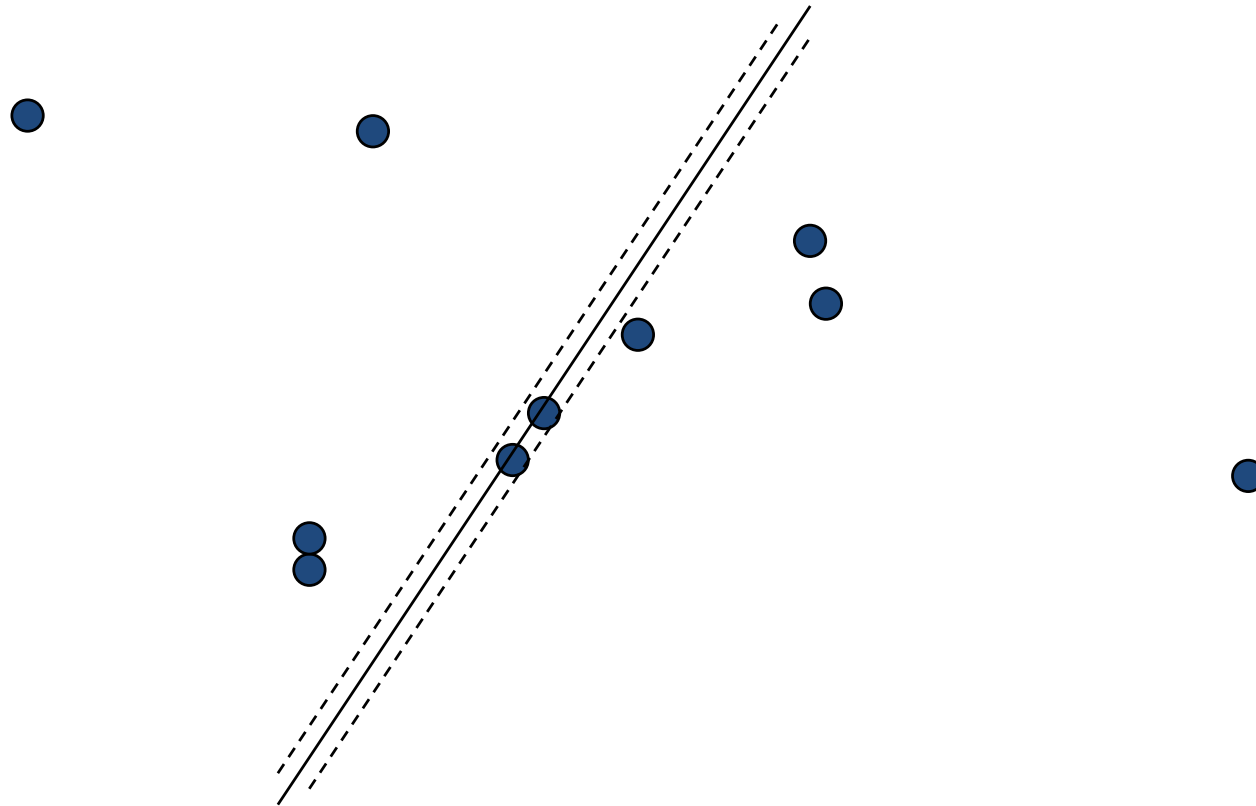
Robust fit of model to data set S which contains outliers

Algorithm

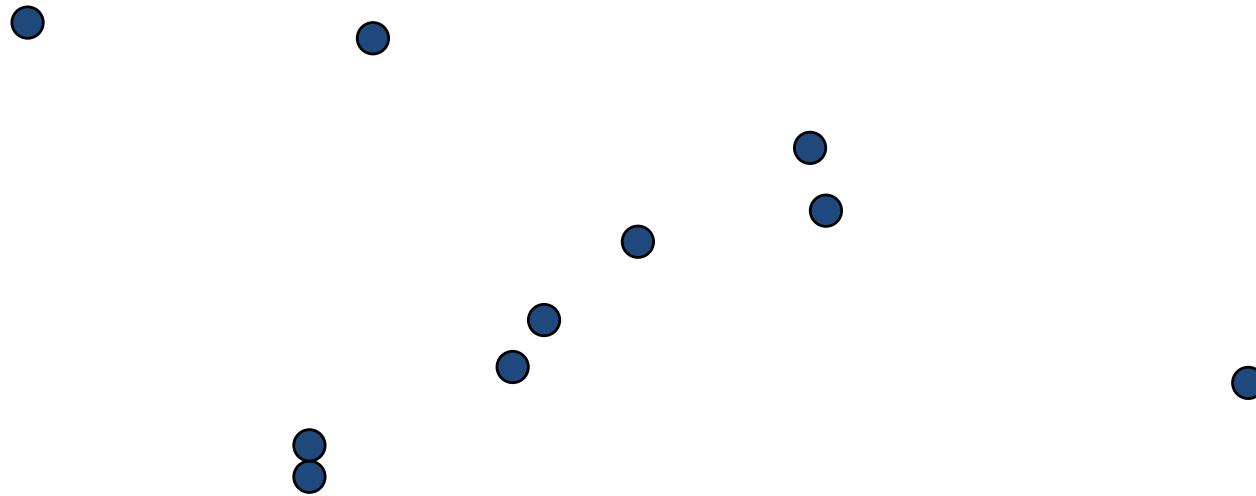
- (i) Randomly select a (minimum number of) sample of s data points from S and instantiate the model from this subset.
- (ii) Determine the set of data points S_i which are within a distance threshold t of the model. The set S_i is the consensus set of samples and defines the inliers of S .
- (iii) If the subset of S_i is greater than some threshold T , re-estimate the model using all the points in S_i and terminate
- (iv) If the size of S_i is less than T , select a new subset and repeat the above.
- (v) After N trials the largest consensus set S_i is selected, and the model is re-estimated using all the points in the subset S_i

(in Hartley and Zisserman, adapted from Fischler '81)

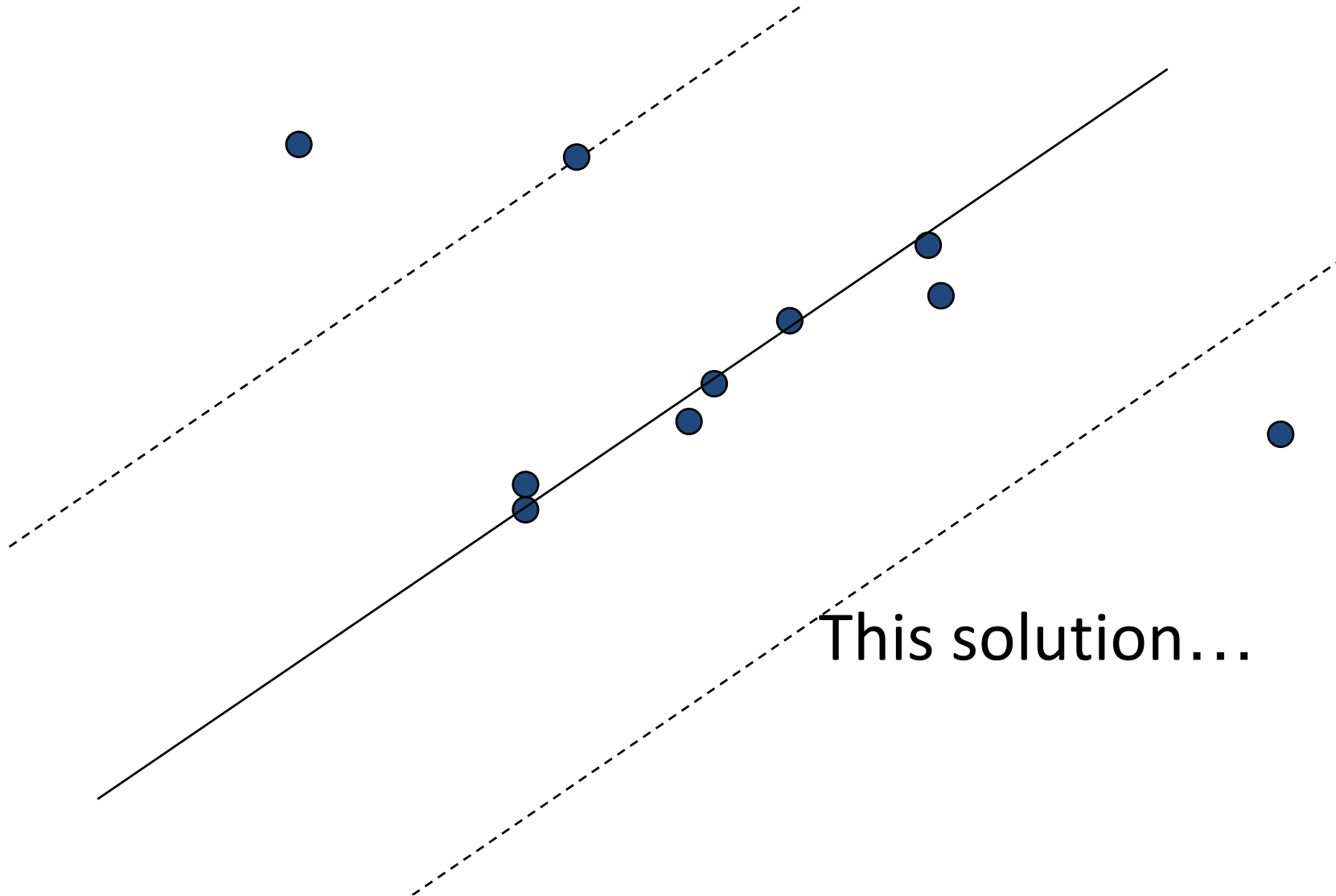
Problem with RANSAC;
threshold too low-no support



Problem with RANSAC;
threshold too high

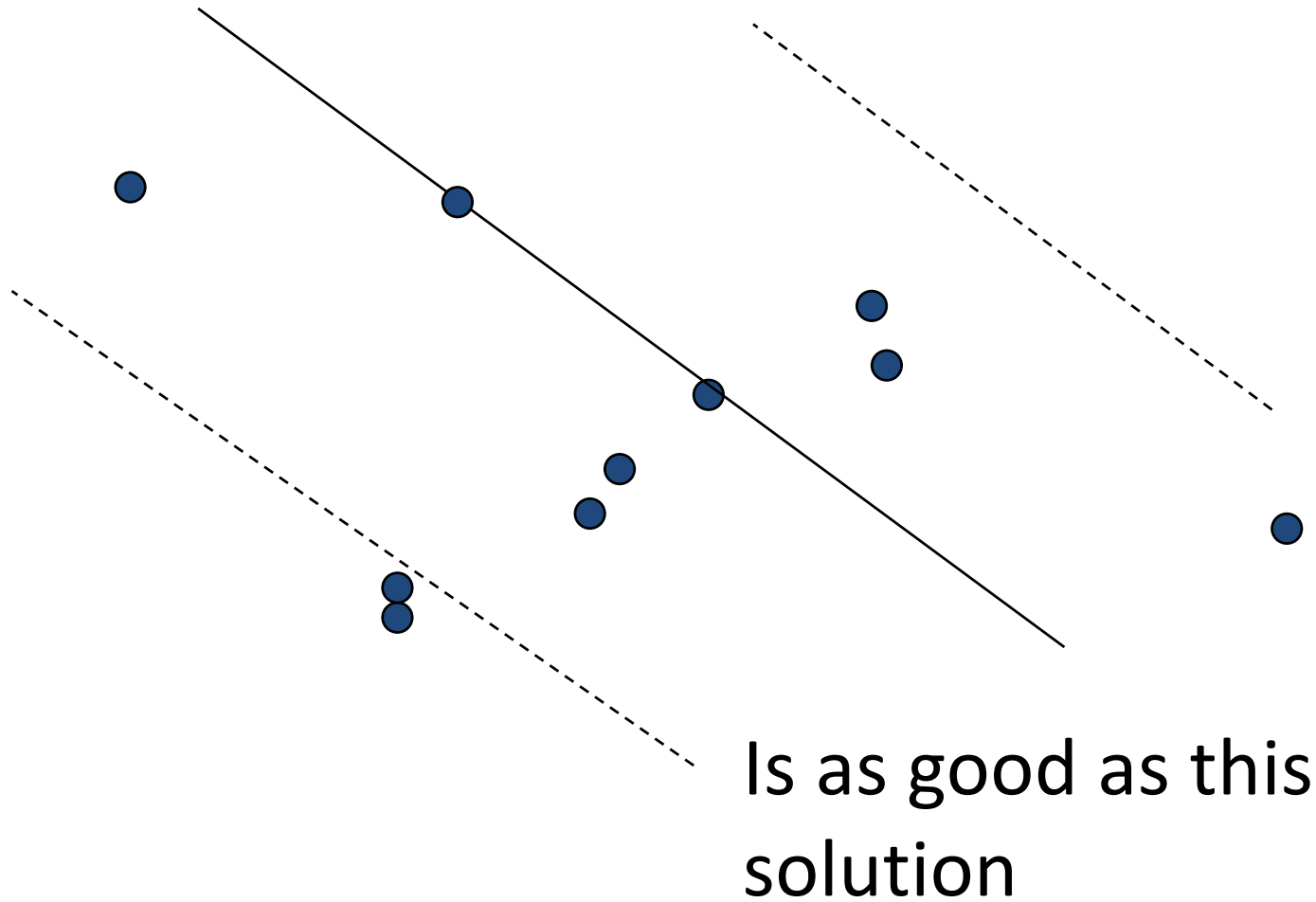


Problem with RANSAC;
threshold too high



This solution...

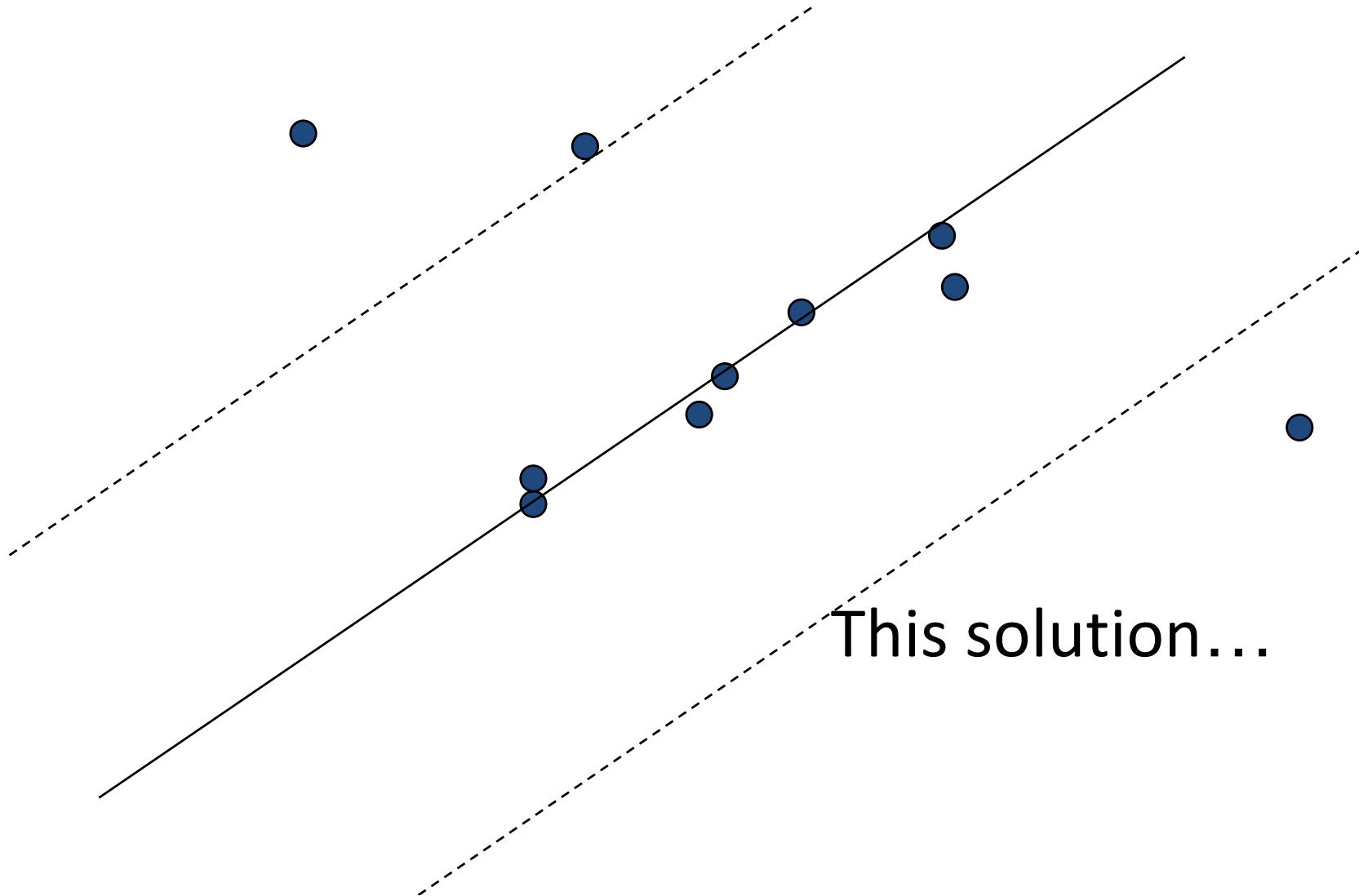
Problem with RANSAC; threshold too high



Cost function

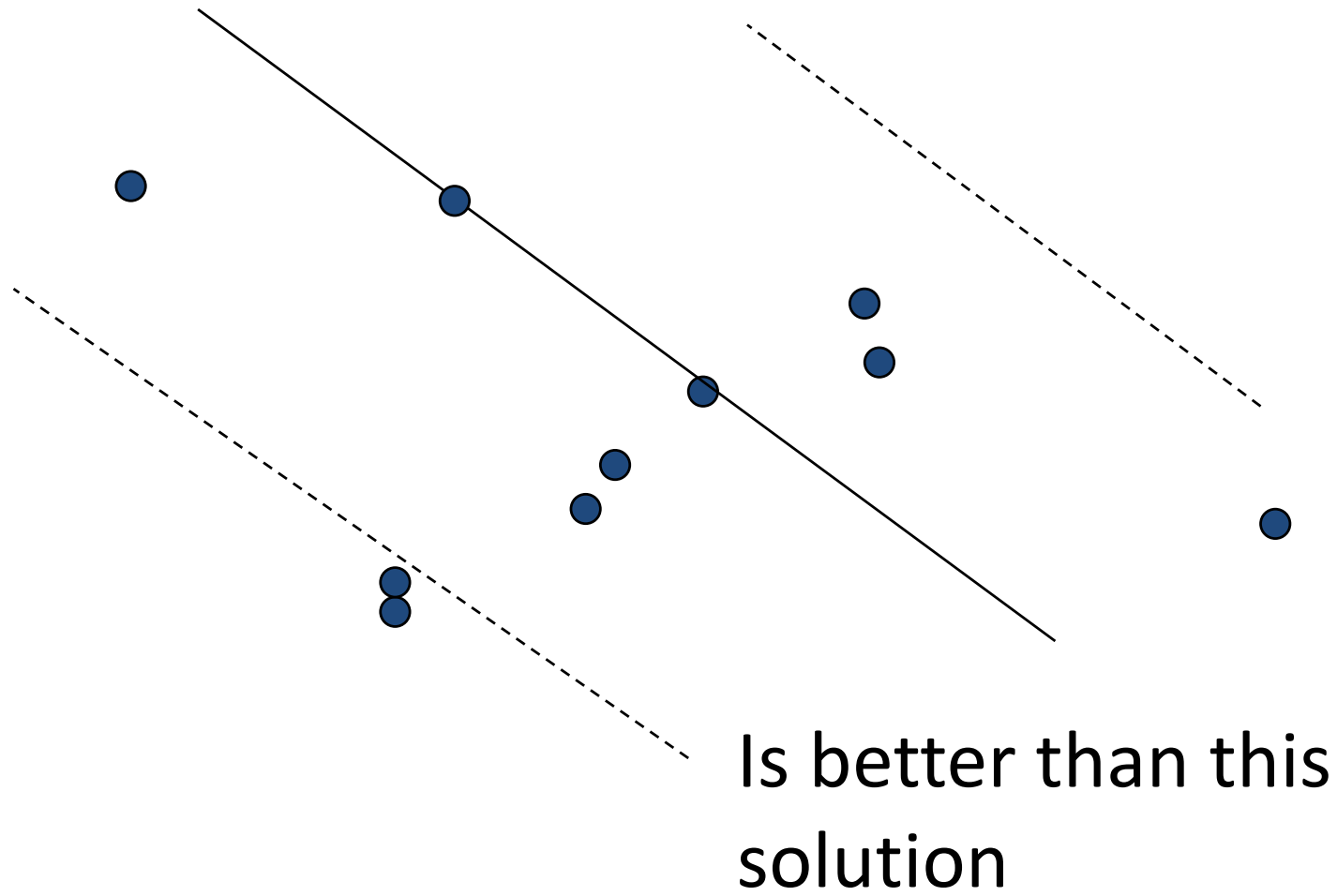
- RANSAC can be vulnerable to the correct choice of the threshold:
 - Too large all hypotheses are ranked equally.
 - Too small leads to an unstable fit.
- The interesting thing is that the same strategy can be followed with any modification of the cost function.

MLESAC



This solution...

MLESAC



Part II: we will visit

Regression => Real-valued output

- Function approximation
- Linear Regression / Least Squares
 - Robust regression (RANSAC to handle outliers)
- Linear regression + regularization
 - Ridge regression
 - The Lasso (a more recent alternative)
- k -NN Regression

Motivation for shrinkage

- Interpretability

- Among a large number of variables X in the model there are generally many that have little (or no) effect on Y
- Leaving these variables in the model makes it harder to see the big picture, i.e. the effect of the “important variables”
- Would be easier to interpret the model by removing unimportant variables (setting the coefficients to zero)

Sample problem: The Credit dataset

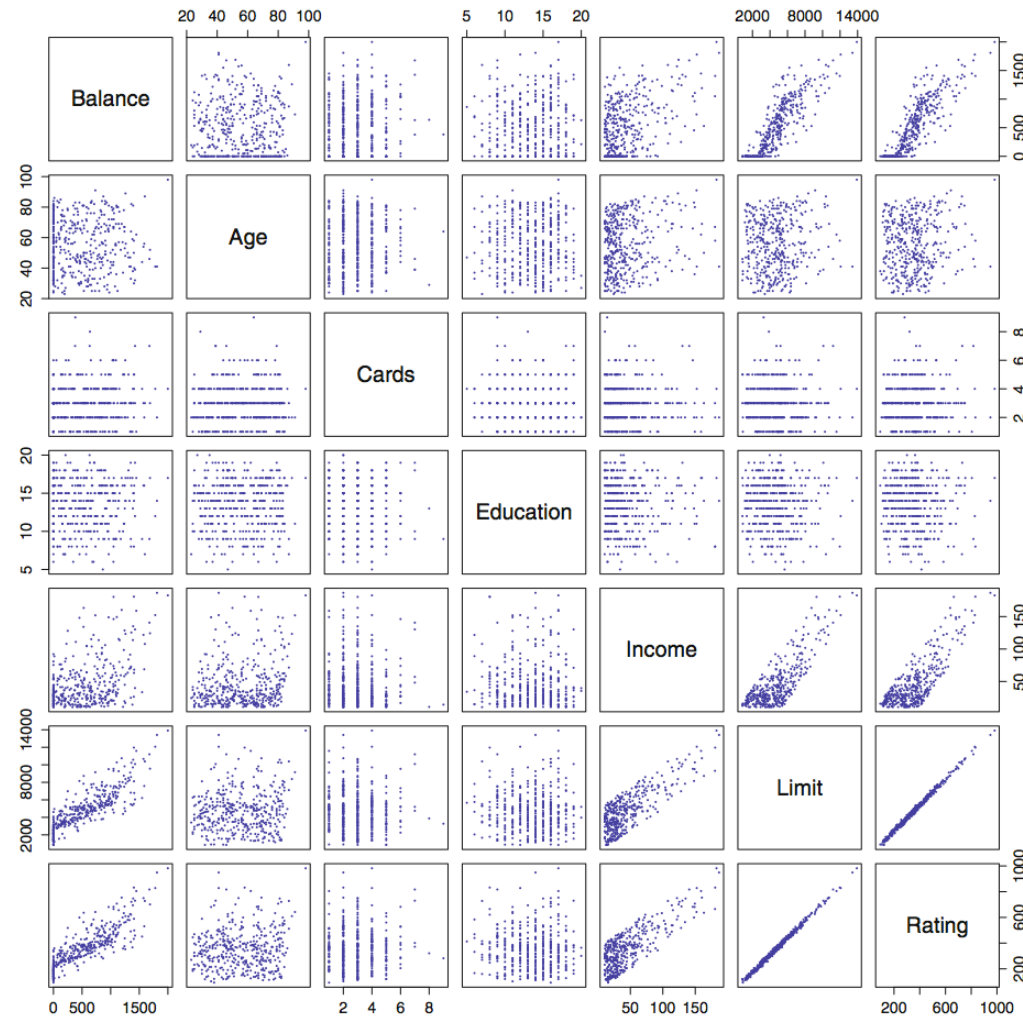


FIGURE 3.6. The **Credit** data set contains information about **balance**, **age**, **cards**, **education**, **income**, **limit**, and **rating** for a number of potential customers.

Figure from An Introduction to Statistical Learning (G. James et al.)

Ridge regression

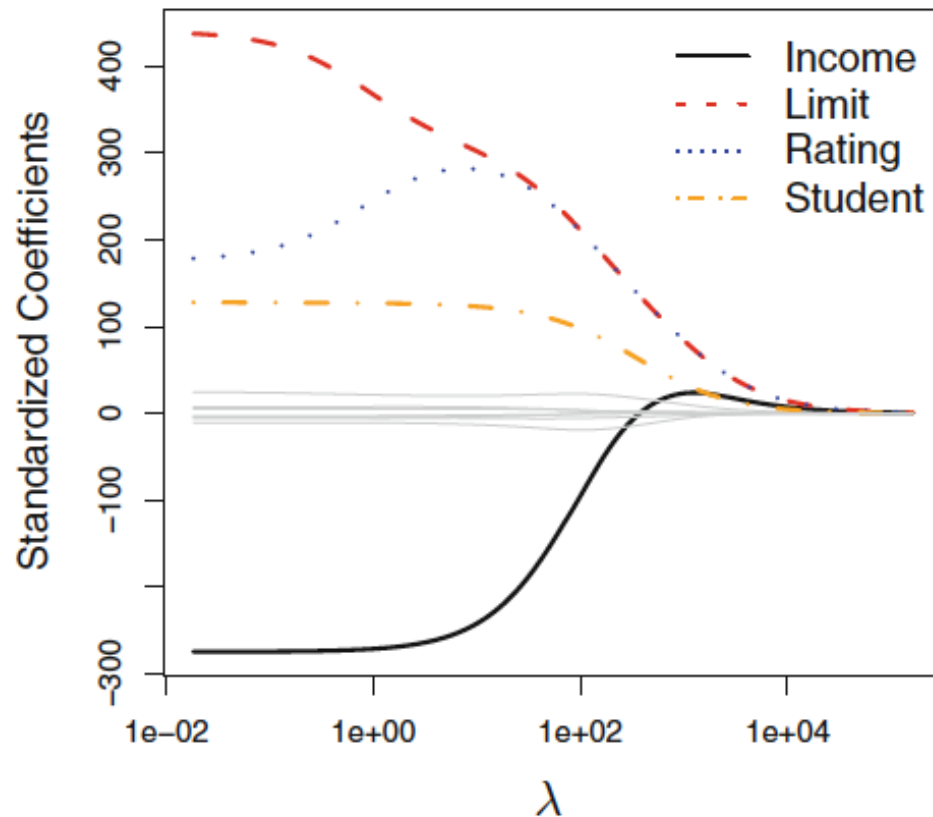
Similar to least squares but minimizes different quantity:

$$RSS + \lambda \sum_{i=1}^d w_i^2$$

The second term is called **shrinkage penalty**

- Shrinkage penalty: small when w_i are close to zero
- The parameter λ : controls the relative impact of the two terms, the selection is critical!

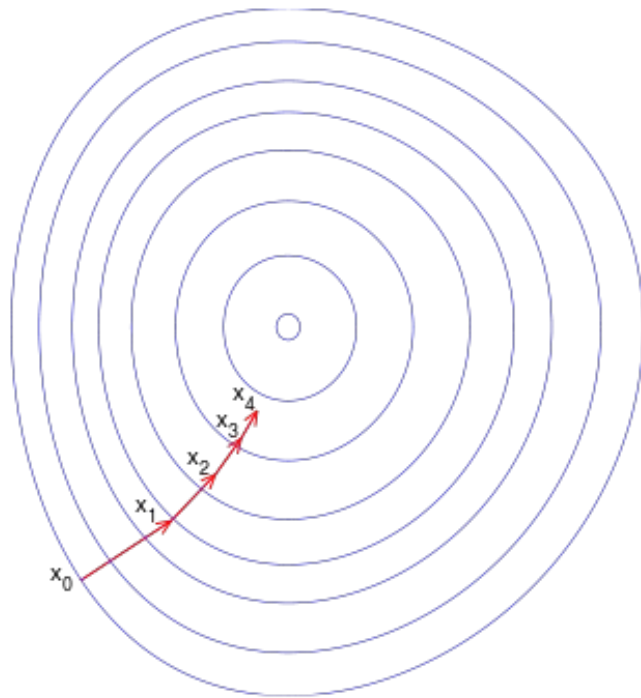
Ridge regression coefficients



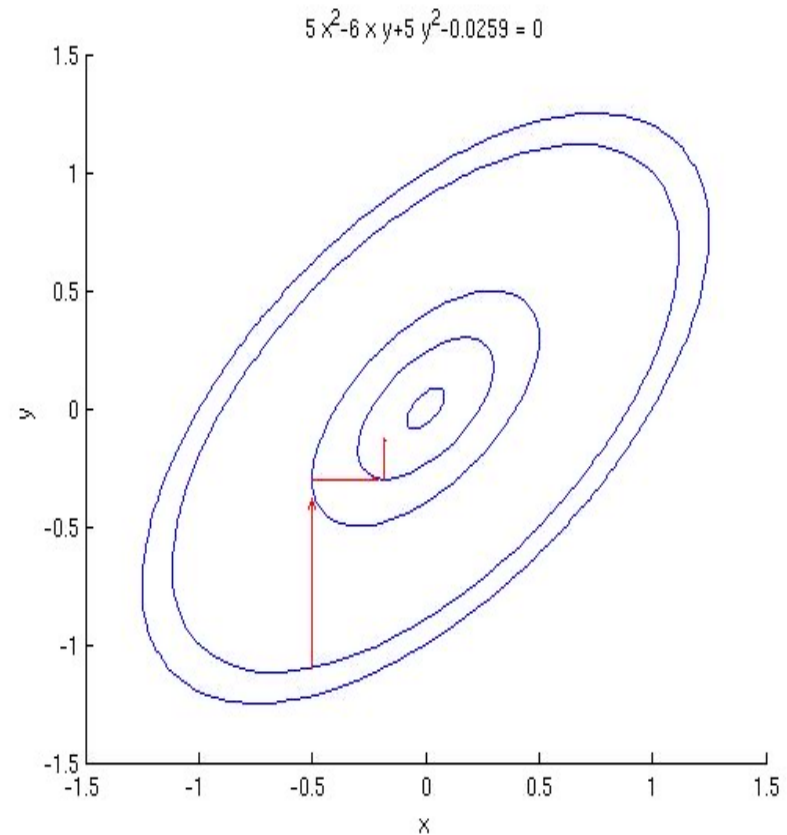
As λ increases, the standardized coefficients shrink **towards zero** (but not exactly forced to zero).

Approaches to parameter estimations

- Gradient decent

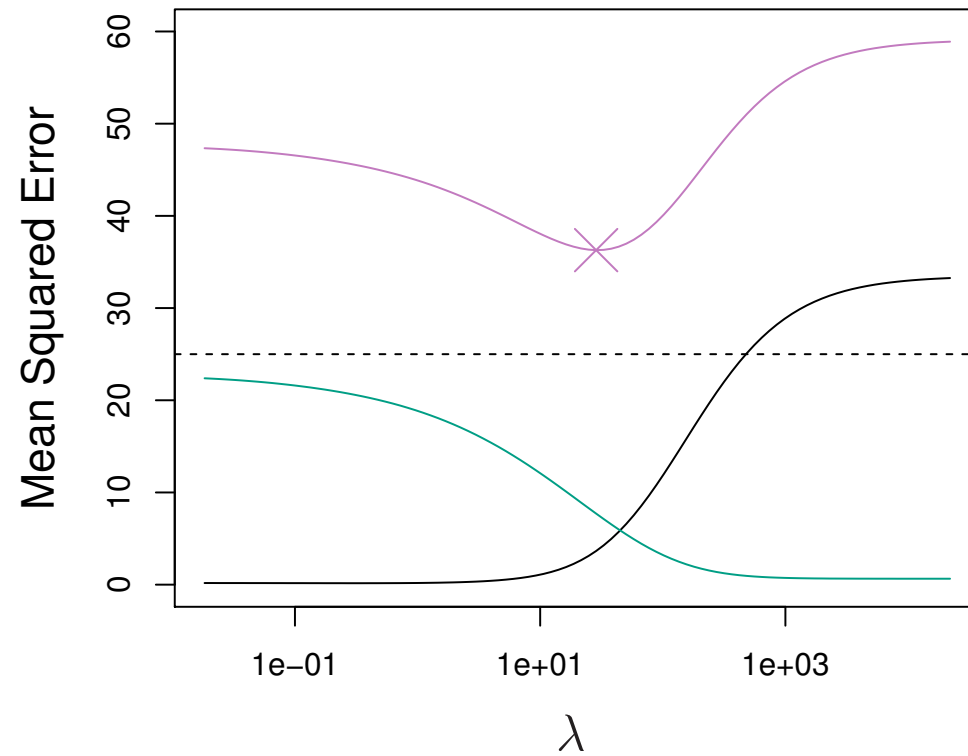


- Coordinate decent



Ridge Regression Bias/Variance

- Green: Variance
- Black: Bias
- Purple: MSE



Increased λ decreases variance while increasing bias

The Lasso

(Least Absolute Shrinkage and Selection Operator)

Similar to ridge regression but with slightly different term:

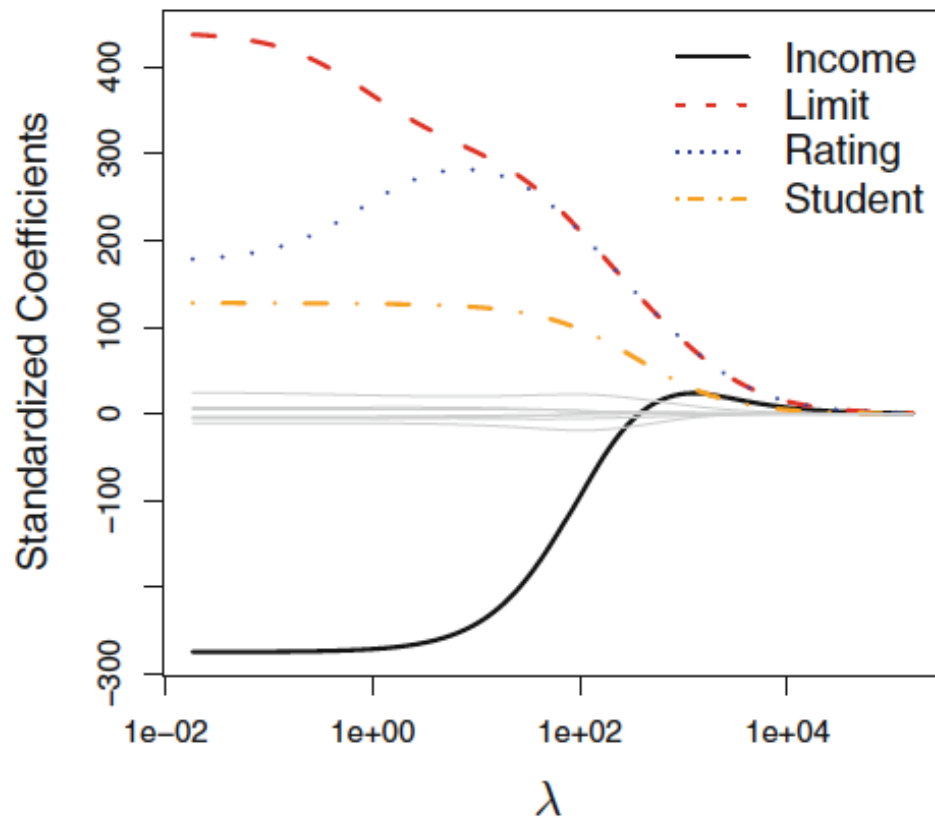
$$RSS + \lambda \sum_{i=1}^d |w_i|$$

The **shrinkage penalty** is now replaced by **l_1 norm**

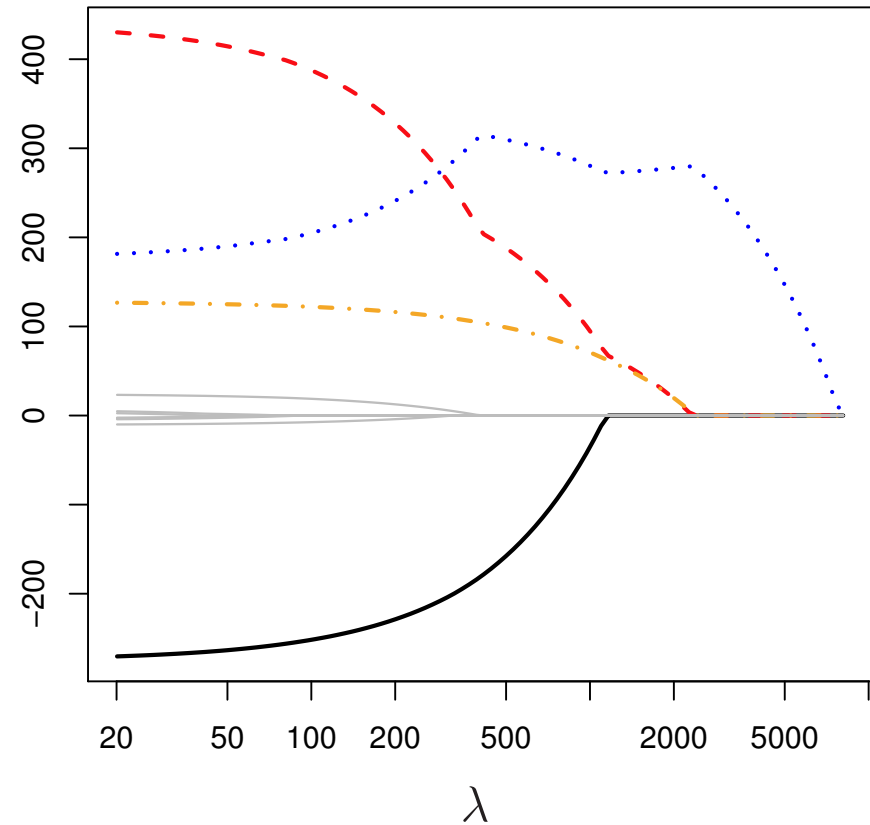
- The lasso could be proven mathematically that some coefficients end up being set to **exactly zero**
 - variable selection
 - yielding sparse model

(whereas Ridge regression includes **all features** in the final model, making it harder to interpret – its drawback.)

Comparison of estimated coefficients



Ridge regression



The Lasso

Another formulations

For every value of λ there is some s such that the equations will give the same coefficient estimates:

- Ridge regression: Minimizing $RSS + \lambda \sum_{i=1}^d w_i^2$

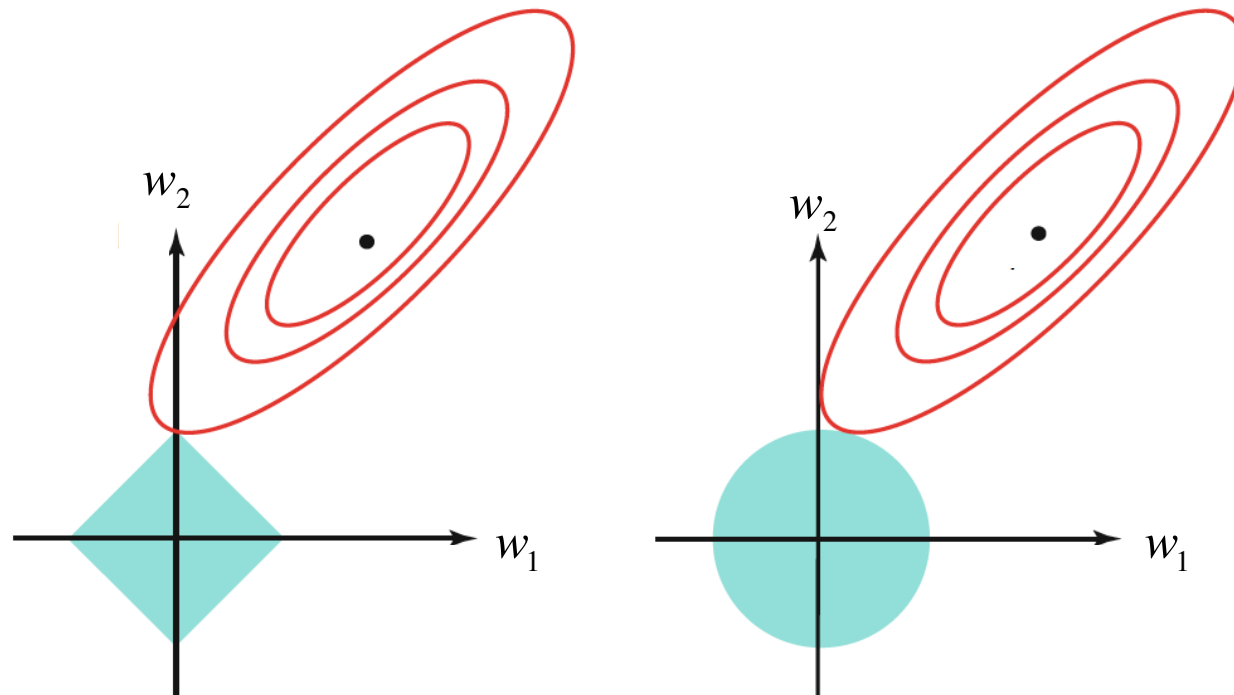
Minimizing $RSS, \text{sub.to } \sum_{i=1}^d w_i^2 \leq s$

- Lasso: Minimizing $RSS + \lambda \sum_{i=1}^d |w_i|$

$RSS, \text{sub.to } \sum_{i=1}^d |w_i| \leq s$

The variable selection property ($d = 2$)

The coefficient estimates: the first point where an ellipse contacts the constraint region as it expands.



The solid blue areas are the constraint regions for
Left: the Lasso
Right: Ridge regression

Part III

- k -NN Regression

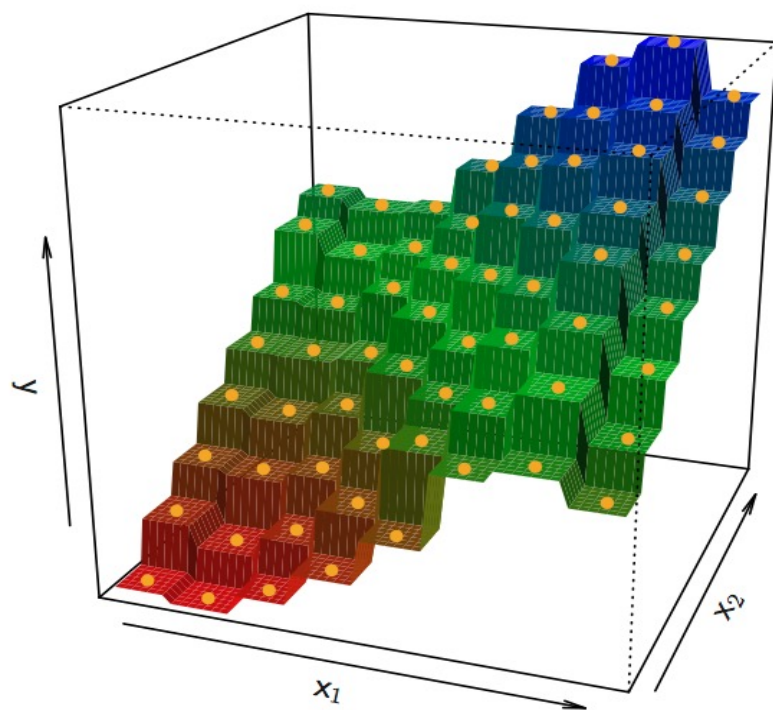
k -NN Regression (non-parametric)

- Similar to the k -NN classifier
- To regress Y for a given value of X , consider k closest points to X in training data and take the average of the responses.

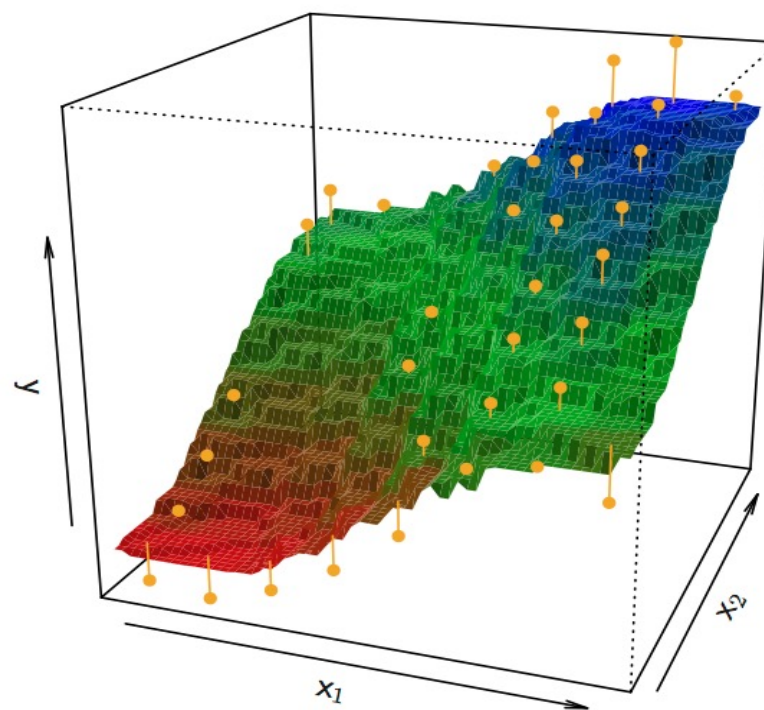
$$f(x) = \frac{1}{k} \sum_{x_i \in N_i} y_i$$

- Larger values of k provide a smoother and less variable fit (lower variance!)

Example plots of $\hat{f}(x)$ with k -NN regression (2d)



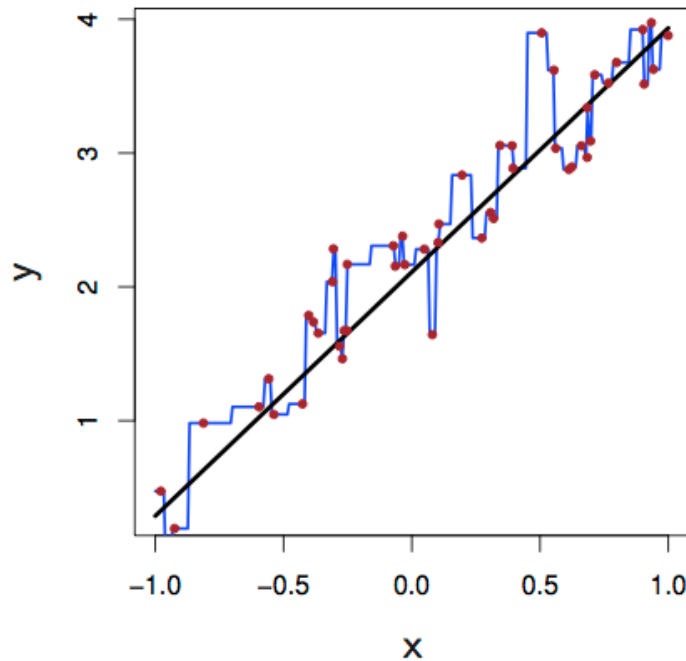
$k = 1$



$k = 9$

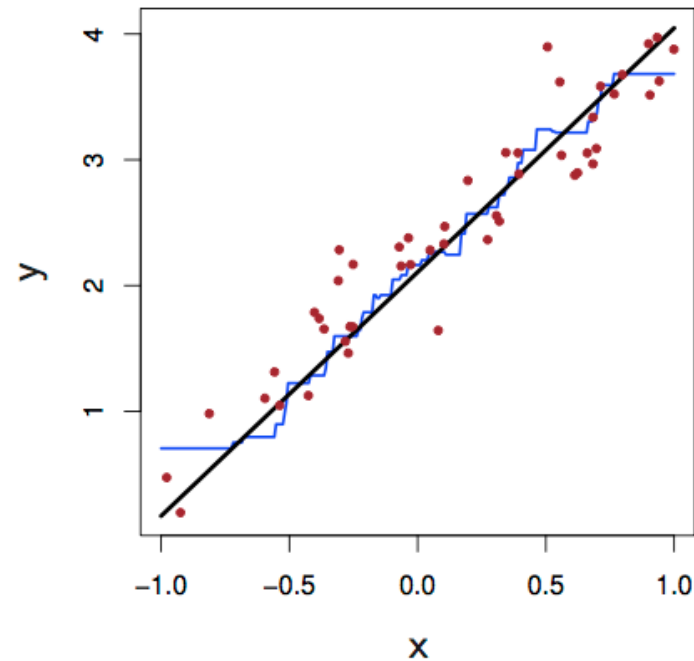
In higher dimensions k -NN often performs worse than linear regression.

Example plots of $\hat{f}(x)$ with k -NN regression (1d)



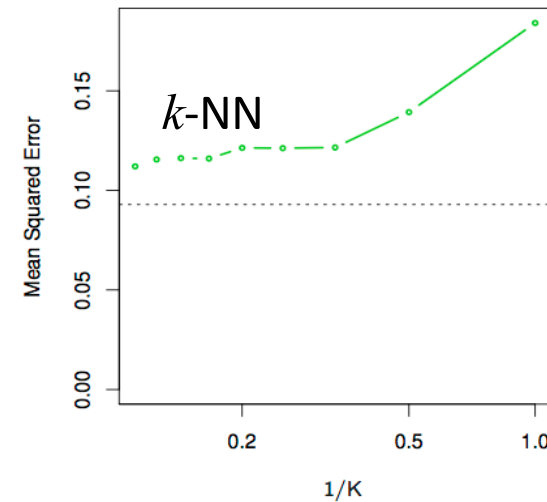
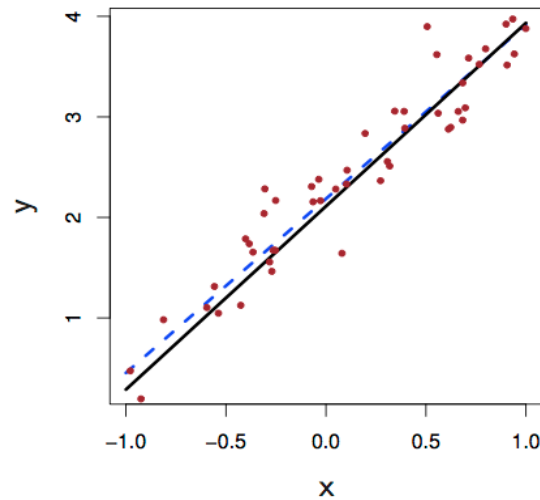
$k = 1$

Black: function
Red: observed
Blue: estimated

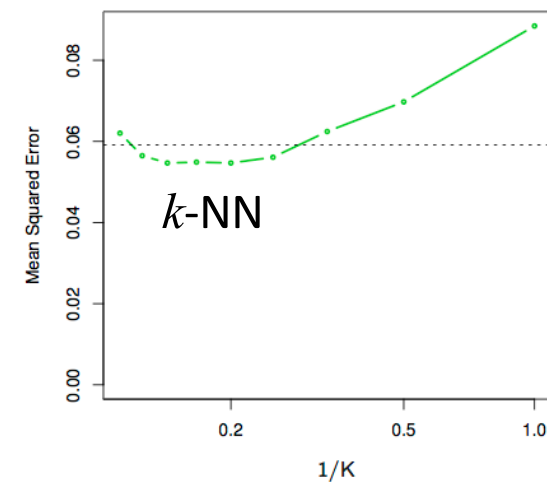
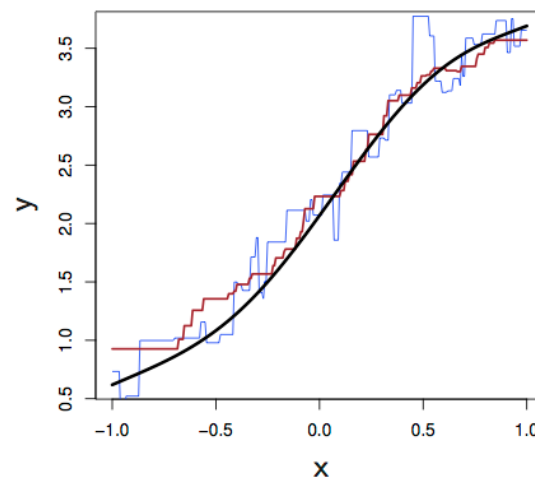


$k = 9$

k -NN vs. Linear Regression (MSE)



← Least square



← Least square

$k = 1$ (blue)
 $k = 9$ (red)

Parametric or Non-parametric?

- How will those compare in what setting?
 - If the parametric form is close to the true form of f , the *parametric* approach will outperform the *non-parametric*
 - As a general rule, *parametric* methods will tend to outperform *non-parametric* when there is a small number of observations per predictor (i.e. in a high dimension).
 - Interpretability stand point: Linear regression preferred to KNN if the test MSEs are similar or slightly lower.